

FL24 Vorobeychik Undergraduate Research

Nasir Sims

December 2024

1 Introduction

My name is Nasir Sims, and I am a second year Computer Science student at Washington University in St. Louis, I have been working with Dr. Vorobeychik and his research team this past semester to further safety advancements in autonomous driving systems. My task has been to design, test, and implement intersection handling in the open-source Carla environment. When the agent approaches an intersection, it should be able to detect it, move to the appropriate lane, and perform the maneuver in a correct manner.

2 Code Basis

The majority of the code was taken from a forked repository of v-e2e-rl-ad created by Owen Ma, but worked on by researchers over the summer as I have been told. This modified version includes numerous changes, including new files such as `alt_global_route_planner.py`, which is the script that provides route planning in CARLA, differing from `route_planner.py` in the former implementation via implementing waypoint generation and decision-making for navigation. `route_planner.py` focuses on a buffer-based, step-by-step waypoint management system with a dynamic computation of waypoints during runtime, `alt_global_route_planner.py` typically leverages a higher-level, graph-based global planning approach that pre-computes routes over the map's topology utilized for intersection detection. This distinction allows `alt_global_route_planner.py` to prioritize global optimality and strategic planning over longer routes, whereas `route_planner.py` emphasizes reactive, localized navigation.

3 First Steps

I had no machine learning experience prior to joining this team, but Owen Ma helped me break down some of the complexities in the work he had put into this project. v-e2e-rl-ad was primarily based on OpenAI's Reinforcement Learning documentation, so I read those articles published by OpenAI.

3.1 LaneNet

My first step was understanding the open-source neural network LaneNet that Owen Ma used for lane marking detection. I thoroughly went through the GitHub Repository, and understood what it did for autonomous driving simulations like the ones we run. Next, I went through the drive folder that we used to train LaneNet from Mini City to understand the scale of the raw training data we used.

3.2 Changes to the codebase

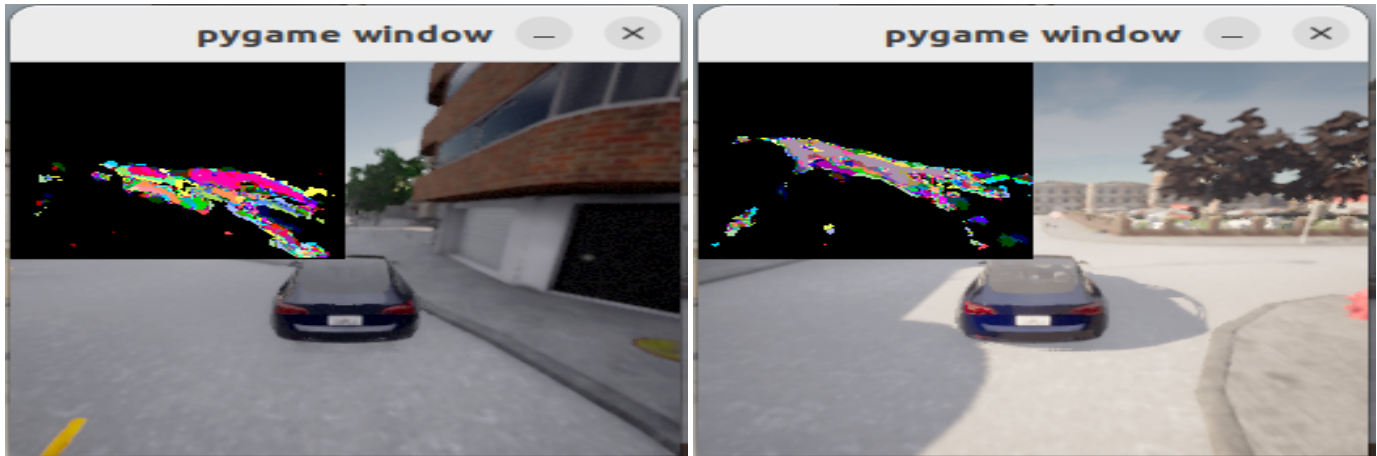
With this new version of v-e2e-rl-ad, there comes additions to the model. The model is LaneNet, but there is a new *left_model_path* and a *right_model_path*, which are

```
/"sys"/LaneNet_lane_detection_pytorch/log/left_turn_model/loss = 0.4402_miou = 0.3000_epoch = 100.pth  
/"sys"/LaneNet_lane_detection_pytorch/log/left_turn_model/loss = 0.4759_miou = 0.2905_epoch = 234.pth
```

respectively.

Inputting these models from the logs seems to have no change, the behavior of the agent is still following that of the LaneNet model.

4 Output



(a) Path into the curb

(b) No intersection behavior

Figure 1: Failures of the Agent (Town02)

As observed by the figures, the agent struggles with intersections (and sharp turns, but that may be a failure with LaneNet model) as it will not behave in the typical manner that the left and right turn states account for in *alt_global_route_planner.py*. Additionally, there is a strange artifact in the pygame window that may have been left in for debugging purposes for LaneNet.

4.1 Optimizations

An issue that frequently occurred was when the agent would approach an intersection (typically in Town10), the detection plus maneuver would be computing intensive, and cause the Pygame window to lag, and the epoch would often end before the agent had the chance to perform an action. Therefore I made changes to the code for additional optimizations.

4.1.1 Optimize turn decision

The turn decision method computes the deviation angle using *math.acos*, which is computationally expensive. It also calculates vector norms that are unnecessary if the vectors are already normalized. Since cv

and `nv` are unit vectors (as CARLA's `get_forward_vector()` returns normalized vectors), I simplified the calculation by removing the norm calculations and removed `math.acos` to compare the dot product.

Original:

```
deviation = math.acos(np.clip(
    np.dot(cv, nv)/(np.linalg.norm(cv)*np.linalg.norm(nv)), -1.0, 1.0))
if deviation < threshold:
    decision = RoadOption.STRAIGHT
```

Optimized:

```
threshold_cosine = math.cos(threshold)

dot_product = np.dot(cv, nv)
if dot_product > threshold_cosine:
    decision = RoadOption.STRAIGHT
```

4.1.2 Optimize Cross Product Calculation

Computing full cross products then extracting the z-component is unnecessary when using 2D vectors, so I simplified this by calculating the z-component of the cross product directly using only the x, y components.

Original:

```
next_cross = np.cross(cv, nv)[2]
```

Optimized

```
next_cross = cv[0] * nv[1] - cv[1] * nv[0]
```

4.1.3 Calculating Waypoint Distance

The `_find_closest_in_list` method calculates distances to all waypoints, which is inefficient because waypoints are already ordered, therefore you can start the search from the last known closest index and break early when the distance starts to increase.

Original

```
def _find_closest_in_list(self, current_waypoint, waypoint_list):
    min_distance = float('inf')
    closest_index = -1
    for i, waypoint in enumerate(waypoint_list):
        distance = waypoint.transform.location.distance(
            current_waypoint.transform.location)
        if distance < min_distance:
            min_distance = distance
            closest_index = i

    return closest_index
```

Optimized

```
def _find_closest_in_list(self, current_waypoint, waypoint_list, last_index=0):
    min_distance = float('inf')
```

```

closest_index = last_index
for i in range(last_index, len(waypoint_list)):
    distance = waypoint_list[i].transform.location.distance(
        current_waypoint.transform.location)
    if distance < min_distance:
        min_distance = distance
        closest_index = i
    else:
        break # Distance is increasing, no need to check further
return closest_index

```

These changes moderately improved performance, but without an FPS counter it was hard to tell. These changes were achieved with the help of ChatGPT.

5 Limitations & Future Steps

5.1 Limitations

On multiple occasions I ran into issues with SSH via my MacBook into the desktop in the PhD office. In those circumstances, sometimes they were able to fix themselves, but I am currently running into a more serious issue, that requires me to utilize the Nano bash editor to make changes to my code on the desktop, hindering progress as it is not exactly an elegant coding environment. I have contacted Anindya about this, and we are working towards a solution.

5.2 Future Steps

I have communicated with Ryan Zhang regarding setting up my own Carla environment on my personal computer, as he was also running into issues, and Owen Ma recommended he do this. This winter break I plan on upgrading my GPU to allow for more memory, and be able to run my own environment based on the v-e2e-rl-ad repository, giving me a head-start on my furthered research for the next semester.

The last session I had with Owen, he recommended I work on creating a classification ML model, with the states being left, right, and straight. He mentioned that by doing this, and implementing key-press to perform the actions, I would be better prepared to see if it works, and training the proper model. I believe I was held back by assuming that the version 2 of v-e2e-rl-ad worked on my summer researchers would be plug-and-play, however it seems it needs a bit more tweaking, plus more understanding on my end.

Once I the intersection handling is a properly trained model in Carla, next semester will hopefully be the time where I can translate that work into the F1tenth vehicle system and test the intersection model in the Mini City.

6 Conclusion

Through this undergraduate research I learned a lot about time management, expectations, and steep learning-curves. A general lack of a clear goal in mind hurt the result of my project at the end of this semester, but now I know that I have the basis and understanding for the work that I need to do next year. I appreciate Dr. Vorobeychik for allowing me the opportunity to work with his talented team, Owen Ma for always being open to answer my many questions and take time to explain things in person, as well as the rest of the researchers who always greeted me with a smile and made me feel welcome despite my age and lack of experience. 2025 will see big things for this research project.

7 References

- [1] Xinhang (Owen) Ma. v-e2e-rl-ad [Computer software]. <https://github.com/catac0mb/v-e2e-rl-ad>
- [2] Xinhang (Owen) Ma. v-e2e-rl-ad [Computer software]. <https://github.com/xhowenma/v-e2e-rl-ad>
- [3] Part 1: Key Concepts in RL - Spinning Up Documentation. <https://spinningup.openai.com/en/latest/spinningup/rlintro.html>